15.071: The AI Edge **Fall 2026**

Team project description

Overview

The projects present opportunities to design, prototype, and pitch AI-powered solutions to real business and policy problems. The goal is to identify a meaningful problem, determine the role of AI to address it, build a solution, evaluate its performance and impact, and communicate its value.

> **Project 1, November 20–Predictive AI.** Develop a predictive AI solution to address a business or policy problem. You can either combine machine learning with tabular data and deep learning with unstructured data into a unified project, or pursue a more ambitious deep learning project using unstructured and potentially multimodal data.
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> **Project 2, December 08–Generative and agentic AI.** Design and prototype an AI-powered application built on LLMs, retrieval-augmented generation, or AI agents.

Projects will be completed in self-formed teams of three (teams of four are allowed but should remain an exception). Teams are encouraged to explore problems connected to their professional experience, personal interests, entrepreneurial ideas, or societal challenges. Projects may address a real organization, a fictional client grounded in a realistic setting, a potential startup, or a broader policy problem. The two projects may address different problems, or a team may develop the same idea throughout the semester as increasingly sophisticated AI capabilities become available. In either case, the goal is to create work that is technically sound and managerially relevant.

From a problem to an AI solution

A strong project begins with a problem and a target stakeholder (e.g., executive, policymaker, investor, operations manager, product leader) and articulates a decision, process, or opportunity that could be improved through AI. The purpose is not merely to build a model with high accuracy or a complex workflow, but to address a problem *through* AI and to demonstrate its practical value.

The scope should be ambitious but focused. Real-world problems are inherently complex; the objective is not to capture every detail of an organization or build a production-ready system. Teams should scope a problem that permits meaningful analysis and evaluation in a few weeks.

Each project should address four questions: (i) *What problem are we solving?* Define the target stakeholder, the current process or decision, and the opportunity for improvement; (ii) *What can AI do?* Acquire appropriate data and design an AI-powered solution; (iii) *Does it work?* Evaluate the solution rigorously against appropriate benchmarks; and (iv) *Does it matter?* Translate technical performance into managerial or societal impact (e.g., revenue, cost, profit, risk, quality).

Creativity is strongly encouraged, but complexity is not an objective in itself. A simple model or application that addresses an important problem and produces actionable value will be more valuable than a technically elaborate system with unclear value. Strong projects combine a meaningful problem, a well-designed AI solution, rigorous evaluation, a compelling case for impact, and clear communication. The ultimate goal is to build something that the team understands deeply, is proud to publish, and would be excited to pitch to a real decision-maker.

Project 1: data, modeling, and evaluation

For Project 1, teams are responsible for acquiring a dataset appropriate for the problem. Publicly available datasets (e.g., Kaggle, UCI repository), APIs, and organizational data for which appropriate permission has been obtained may be used. Projects intended for public portfolios must not contain confidential, proprietary, personally identifiable, or otherwise restricted information.

The core of the project is the development and evaluation of a predictive AI solution. Teams should translate the business or policy problem into a well-defined prediction task, identify appropriate inputs and outcomes, prepare the data for modeling, perform rigorous model selection and validation, and establish meaningful comparisons against relevant benchmarks (e.g., predictive baseline, business rule, alternative modeling approaches). Beyond technical performance, teams should assess the practical impact of the solution by quantifying how improved predictions translate into business or policy improvements such as increased revenue, reduced costs, increased efficiency, better service, or reduced risk. Appropriate visualizations are encouraged to showcase impact.

Sample topics are provided below, but you are encouraged to explore topics based on your interests.

- Real estate valuation: Use historical records or emerging data (e.g., photographs) to assess property value and support acquisition, investment, inspection, or insurance decisions.
- Employee attrition and workforce planning: Use employee records, engagement surveys, or employee feedback to predict departures reduce turnover and improve workforce planning.
- Insurance claims: Use records, claim descriptions or photographs to predict claim likelihood or severity and improve pricing, reserving, or case prioritization.
- Hospitality demand and customer experience: Use booking histories, customer characteristics, and reviews to forecast demand and improve pricing, staffing, and service decisions.
- Predictive maintenance: Use equipment sensor, telemetry data and maintenance reports to anticipate failures and improve maintenance planning.
- Retail and e-commerce: Use transaction histories, product attributes, images, or customer behavior to predict demand, classify products, recommend items, or improve merchandising.
- Healthcare operations: Use patient characteristics, measurements, medical images, or clinical notes to predict outcomes and support diagnosis, triage, resource allocation, or delivery.
- Manufacturing quality control: Use production data, product images and sensor data to detect defects, predict quality problems, and improve inspection or production decisions.
- Transportation and mobility: Use trip histories, operational data, images, or other sensor data to predict demand, delays, incidents, or service quality and improve planning and operations.
- Agricultural monitoring: Use historical production and weather data along with satellite or aerial imagery to predict yields, assess land use, and support agricultural decisions.

Project 2: building, evaluating, and deploying an agentic AI application

For Project 2, teams are responsible for designing and prototyping an agentic AI application that addresses a meaningful business or policy problem. Teams should identify the knowledge, data,

tools, and external systems required by the application and design an appropriate workflow. The resulting prototype should be functional and demonstrate the solution's capabilities.

The core of the project is the development and evaluation of an end-to-end agentic AI application. Teams should translate the business or policy problem into a well-defined set of tasks, design the application architecture and workflow, and make thoughtful choices about models, prompts, knowledge sources, tools, and agent behavior. The resulting system should be rigorously evaluated via sample inputs and outputs illustrating both successful outcomes and failure modes. Beyond technical performance, teams should evaluate how workflow augmentation can deliver practical impact, as well as the pathways toward implementation, adoption, and change management.

Again, the objective is not simply to build an impressive demonstration but to demonstrate a clear understanding of the problem, the role and design of the agentic AI solution, the value created relative to the existing process, the requirements for deployment and adoption, and the limitations and risks. An honest analysis of an imperfect system, supported by thoughtful diagnosis and clear recommendations, is more valuable than a technically impressive model with erratic behavior.

- Investment intelligence: Monitor news, identify developments, produce targeted briefings.
- Clinical documentation: Process consultations, update medical records, flag escalations.
- Email: Prioritize messages, summarize information, draft responses, manage follow-ups.
- Expenses: Process receipts, categorize expenses, check compliance, prepare reports.
- Real estate: Search properties, assess market, evaluate investments, recommend acquisitions.

Deliverables

Each project should communicate to both managerial and technical audiences. The video and presentation should answer the managerial questions of *why* the solution matters. The technical repository should answer the technical questions of *how* the solution works.

- **Two-minute video.** A self-contained demo or pitch addressed to the target stakeholder.
- **Presentation (5–10 slides).** A concise deck describing the problem, the AI solution, business impact, and recommendations. For Project 1, the AI solution will focus on the data, the modeling approach, and the key findings; for Project 2, it will focus on the agentic workflow. The presentation should also be addressed to the target stakeholder.
- **Technical repository.** A GitHub repository containing technical components and a one-page `README`, addressed to a data scientist or machine learning engineer in the target organization. For Project 1, this will include executable code and instructions for reproducing the results; for Project 2, this will include the application code or exported workflow, together with a description of the system architecture, setup instructions, and usage.

We encourage projects to be published in professional online portfolios, creating a tangible showcase of each team's work at the interface of AI and management. The portfolio is intended to live beyond the course and may be shared with employers, collaborators, clients, or investors. A selection of projects will be showcased during **Demo Day on December 9**.